

No. of Functional Features Included in the Solution

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the HDI Predictor solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Country Dropdown	Dynamically populated dropdown listing 53 countries from hdi_dataset.csv using a Jinja2 for loop.	Frontend (index.html)	Done	1
2	Input Form Validation	All four input fields use required attribute and type='number' to enforce client-side validation before submission.	Frontend (index.html)	Done	1
3	Dataset Loading with File Guard	load_dataset() reads hdi_dataset.csv using pandas. Checks file existence with os.path.exists() and raises a descriptive FileNotFoundError if missing.	Backend (app.py)	Done	1
4	ML Model Training	train_model() isolates 4 feature columns (X) and hdi_score target (y), then trains RandomForestRegressor with n_estimators=300 and random_state=42.	ML Engine (app.py)	Done	1
5	HDI Score Prediction	POST route extracts form values, casts them to float inside try-except, builds a single-row DataFrame, and runs model.predict() to get a decimal HDI score.	Backend (app.py)	Done	1
6	Tier Classification	classify_hdi() maps the predicted score to a tier label — Very High (≥ 0.800), High (≥ 0.700), Medium (≥ 0.550), Low (< 0.550) — using UNDP official thresholds.	Backend (app.py)	Done	1
7	Dynamic Result Display	Jinja2 {% if result %} block conditionally renders the predicted tier and rounded score in a styled result box only after a successful POST request.	Frontend (index.html)	Done	1
8	Production Deployment	App configured with host='0.0.0.0' and dynamic PORT env variable. Deployed live on Render using gunicorn as the WSGI server.	Deployment (Render)	Done	1

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (Dataset load, Model train, Prediction, Classification, Form validation, Result display)
Additional / Nice-to-Have Features	2 (Country dropdown, Production deployment)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Country dropdown, Input form validation, Conditional result display
2	Backend / Logic	3	Dataset loading with file guard, HDI score prediction, Tier classification
3	Database / Storage	1	CSV dataset (hdi_dataset.csv) with 53 countries and 7 columns
4	ML / Integration	1	RandomForestRegressor model training and inference via scikit-learn
5	Security / Deployment	1	try-except error handling, Gunicorn WSGI, Render production deployment